A Robust Canonical Correlation Analysis of Digital Distraction and Deep Learning Approach Among University Students

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Abstract **- This research examines the multivariate relationship between digital distraction and deep learning approach among university students using the Robust Canonical Correlation Analysis (RCCA) method in conjunction with the Minimum Covariance Determinant (MCD) estimator. A questionnaire was administered towards university students, with a final sample total of 107 respondents. It measured three dimensions of Digital Distraction, being Social Media Urge, Digital Notification Sensitivity, and Attention Fragmentation alongside with three dimensions of Deep Learning Approach, being Critical Thinking, Conceptual Understanding, and Reflective Learning. Since the data used failed the multivariate normality assumption based on Mardia's test, the more robust approach (MCD) was applied to produce estimates that are unbiased and resistant towards outliers. Results indicate that the first canonical function explains 41.94% of the shared variance between the two constructs / main variables. Within the digital distraction profile, Attention Fragmentation is deemed as the most influential factor with a loading score of 0.9781, while Reflective Learning is deemed the highest outcome loading on the deep learning side with a loading score of 0.9039. The high reflective learning loading indicates that students may overestimate their ability to learn deeply despite digital distractions, which is consistent with metacognitive bias and goes against the first proposed negative link. These results show that students learn more effectively and efficiently when they are less distracted by digital multitasking and are helped to develop strong self-regulated learning habits and better awareness of how they learn.**

Keywords **- Digital Distraction, Deep Learning Approach, Robust Canonical Correlation Analysis, Minimum Covariance Determinant**

1. INTRODUCTION

1.1. Background

The way students learn today is very different from the past because of the advances in digital technologies. Digital devices are now heavily relied on to access information, communicate and do assignments. These digital devices bring lots of benefits, but they also bring difficulties such as digital distraction (Martin et al., 2025).

When digital devices start to disrupt learning activities, that's when it stops being beneficial and starts being harmful. Checking notifications, replying to messages, and multitasking are examples of harmful habits that can happen because of digital devices. It's even more difficult now to prevent digital distraction because of the increasing incorporation of technology to education. As a result, students struggle to focus during study sessions. (Wolff, 2024).

Recent studies show that digital distractions have negative effects on a student's learning performance. Cognitive processes are less effective when their focus is split between digital activities and academic activities. Attention Fragmentation is a condition where the capacity to understand and process concepts in depth (Pérez-Juárez et al., 2024). Interruptions and distractions increase cognitive load, making students struggle even more to engage in complex thinking processes like analysis and evaluation (Amsiyah, 2026).

Digital distraction is closely related to self-regulated learning, which is a key element of deep learning approach. When distractions happen frequently, students have less control over their learning process, ability to organize, monitor, and evaluate the way they are learning thus resulting in ineffective learning and inability to improve academic performance (Zayed, 2025).

The deep learning technique relies heavily on comprehension, critical thinking, and reflection. Students

who use this approach do not only memorize but understand the meaning and its significance. They are more likely to use the knowledge in different situations, connect ideas, and have better critical thinking. Digital distraction is holding the effectiveness of the deep learning approach back (Flanigan, 2025).

Given this issue, it is important to use a statistical strategy to thoroughly investigate the connection between digital distraction and the deep learning approach. This study examines the link between several variables from both domains using Canonical Correlation Analysis (CCA). However, traditional CCA are extremely sensitive because of multivariate outliers from distributional assumptions. A robust method using the Minimum Covariance Determinant (MCD) overcomes this constraint and provides stable calculations. Robust Canonical Correlation Analysis (RCCA) is unaffected by anomalies which may show in the data structure, which makes precisely capturing the relationship between digital distraction and deep learning techniques possible.

1.2. Problem Foundations

Based on the background given, a few problems could be identified,

a. Is there a significant relationship between digital distraction and deep learning approach?
b. Which variables contribute the most to the relationship between digital distraction and deep learning?
c. How effective is the Robust Canonical Correlation Analysis method using the MCD approach?

1.3. Objectives

a. To analyze the relationship between digital distraction and deep learning approach
b. To identify the most influential variables in both constructs
c. To apply the Robust Canonical Correlation Analysis method using the MCD approach

1.4. Benefits

This study produces both theoretical and practical contributions. From a theoretical perspective, this study clarifies the connection between digital distraction and the deep learning approach through the use of a reliable statistical method, enhancing knowledge in educational psychology and multivariate statistics. This study adds to the collection of research on effects of digital behavior on student learning outcomes and cognitive processes.

The results of this study can help educators understand the impact digital distraction has on learning quality, allowing them to create an environment to boost students' learning process as well as a teaching method that works efficiently. In addition, this study provides insights that could help students be more aware towards learning habits and improve their self-regulated learning practices, which is important to enhance focus and critical thinking (Wang, 2025).

2. THEORETICAL STUDY

2.1. Digital Distraction

Digital distraction means interruptions to our focus caused by using digital devices while engaging in educational activities. Checking notifications, answering texts, and/or multitasking are examples of these interruptions. Digital distraction has been growing more and more significant due to the ever growing integration of digital technology to learning activities (Martin et al., 2025).

Cognitive performance can be negatively affected by digital distraction. Splitting time between academic activities and online diversions greatly reduces concentration and focus. When the brain is forced to switch tasks simultaneously it leads to lower productivity and error prone, this is called attention fragmentation. This causes students to find it even more difficult to stay focused and renders them unable to fully understand the subject matter (Wolff, 2024).

Digital distraction also adds cognitive load, which refers to the amount of effort needed to process information. Constant interruptions caused by digital distraction results in a heavier demand on working memory, which as a result leads to students having difficulties to process information and decreasing their ability to analyze and evaluate (Amsiyah, 2026).

Furthermore, digital distraction is closely related to behavioral patterns such as impulsive checking and habitual media use. These behaviors reduce student's ability to stay focused on long tasks and negatively impact learning outcomes. Studies also show that students with higher levels of digital distraction tend to have lower

academic performance and reduced engagement in learning activities (Pérez Juárez et al., 2024).

2.2. Deep Learning Approach

Deep learning approach is a learning strategy where students aim to understand the meaning of the material rather than simply memorizing it. This approach focuses on critical thinking, conceptual understanding, and reflective learning.

Critical thinking involves the ability to analyze information, evaluate arguments, and make logical conclusions. Connecting ideas and understanding relationships between concepts is the meaning of conceptual understanding. Reflective learning is evaluating your own learning method to then improve it.

Deep learning approach users tend to be active in their education, more likely to ask questions, consider different perspectives, and apply their knowledge. This results in a deeper understanding and longer retention of the information.

Self-regulated learning and the deep learning approach are closely linked. The ability to control their learning process, set goals, track progress, and modify strategy is referred to as self-regulated learning. Self-regulation helps students stay focused, resulting in more effective learning and deeper understanding of the material. (Wang, 2025).

2.3. Relationship Between Digital Distraction and Deep Learning

In general, there is a negative correlation between the deep learning approach and digital distraction. Students' ability to concentrate, which is crucial for deep learning processes, is diminished by digital distraction. Students are less able to participate in critical thinking and conceptual knowledge when they are constantly distracted by digital devices. Their inability to fully digest information is a result of their split attention. Because of this, individuals frequently use surface-level learning techniques like memorization rather than deep comprehension.

Research shows that digital distraction is negatively associated with self regulated learning. Students who experience higher levels of distraction have lower ability to plan, monitor, and evaluate their learning process (Zayed, 2025). This directly affects their ability to engage in deep learning.In addition, digital distraction increases cognitive load and reduces engagement, making it difficult for students to reflect on their learning. Reflection is an important component of deep learning, as it allows students to improve their learning strategies and achieve better outcomes (Flanigan, 2025).

Based on existing literatures, digital distraction is generally expected to have a negative influence towards deep learning approaches. However, empirical findings may reveal more subtle patterns depending on how students perceive their own capability of learning.

2.4. Canonical Correlation Analysis (CCA)

Canonical Correlation Analysis is a multivariate statistical technique used to analyze the relationship between two sets of variables. Unlike simple correlation, which examines the relationship between two variables, CCA analyzes multiple variables simultaneously. The main objective of CCA is to find linear combinations of variables from each set that have the highest correlation. These combinations are called canonical variates. The strength of the relationship is measured using canonical correlation coefficients.

CCA is particularly useful in this study because both digital distraction and deep learning approach consist of multiple indicators. By using CCA, it is possible to understand the relationship between the two constructs more comprehensively.

2.5. Robust Canonical Correlation Analysis with MCD Approach

In classical statistical methods, one important assumption is that the data follows a multivariate normal distribution. However, in real world data, this assumption is often violated due to the presence of outliers or non normal distributions. When the assumption of normality is not met, classical CCA may produce biased and unreliable results. Therefore, robust statistical methods are needed to handle such conditions.

The Minimum Covariance Determinant (MCD) method is a robust technique used to estimate covariance matrices. This method works by selecting a subset of the data that minimizes the determinant of the covariance matrix, thereby reducing the influence of outliers. The covariance matrix becomes more stable when using the MCD approach. This leads to a more reliable and accurate result when using the Canonical Correlation Analysis.

3. METHODOLOGY

3.1. Research Design

This study uses Robust Canonical Correlation Analysis (RCCA) with the Minimum Covariance Determinant (MCD) estimation point to analyze the relationship between both digital distraction and deep learning approach among the university students. This method was selected because the data of the survey do not always follow the ideal statistical assumptions and may contain responses which are very extreme from some participants. These responses can directly affect the results of standard Canonical Correlation Analysis (CCA), by making the relationships between variables either stronger or weaker than they actually do. The MCD estimation point is used to solve this issue by reducing the influence of outliers and producing more stable estimates. This will allow the analysis to provide a clearer impression of how digital distraction is related to students deep learning approach.

3.2. Population and Sample

The population of this study consisted of undergraduate students who regularly use digital devices in their academic activities. A purposive sampling technique was used to select those respondents who were considered relevant to the objectives of the study. Participants were required to have experience using digital technologies, for example smartphones, laptops, and online learning platforms like zoom, google meet, etc as part of their learning process. The data were collected by an online questionnaire, with the result in 107 valid responses. This sample size was considered adequate for CCA because of its minimum requirement for multivariate statistical procedures involving the six observer variables.

3.3. Variables and Measurement

The variables in this study are divided into two sets:

Digital Distraction (Variable X):

- X1 (Social Media Urge): Students' tendency to switch between academic work and non-academic digital activity.
- X2 (Digital Notification Sensitivity): assesses how obsessively students react to texts, alerts, and social media posts while they are studying.
- X3 (Attention Fragmentation): Measures the habit of simultaneously engaging with multiple media streams or digital applications during learning activities.

Deep Learning Approach (Variable Y):

- Y1 (Critical Thinking): evaluates the ability to examine, challenge, and assess academic knowledge instead than taking it at face value.
- Y2 (Conceptual Understanding): evaluates how well students develop conceptual understanding by relating new ideas to existing knowledge and practical experiences.
- Y3 (Reflective Learning): Evaluates the habit of self-monitoring, reviewing learning outcomes, and synthesizing diverse ideas into a coherent framework.

Each variable is measured using a Likert scale (1-6), where respondents indicate their level of agreement. This scale is used to get subjective perceptions in a quantitative form.

3.4. Data Collection Method

The data were gathered by using a structured online questionnaire which is administered to undergraduate students. The instrument was developed based on the dimension of both variables digital distraction and deep learning approach and measured using a six point likert scale. Before proceeding to the main analysis, validity and reliability testing were performed to ensure the quality and also consistency of the data collected.

3.5. Data Analysis Procedure

The data analysis is conducted in several stages:

1. Validity and Reliability testing
2. Normality Assumption and Multicollinearity testing
3. Robust Covariance Estimation using the MCD approach
4. Robust Canonical Correlation Analysis
5. Interpretation of results

3.6. Validity and Reliability Testing

To ensure that every value measures correctly, validity tests were run. Correlation analysis were done to achieve this. To evaluate the consistency, reliability testing was run. When used under the correct circumstances, this reliable method results in stable and consistent findings.

3.7. Assumption Testing

Before conducting Canonical Correlation Analysis, several assumptions were checked. The Mardia test was used to assess multivariate normality. Multicollinearity was also examined to ensure that the variables were not too highly correlated, which could interfere with the analysis.

3.8. Robust Covariance Estimation (MCD Method)

By the results of the multivariate normality test, a robust covariance matrix was estimated using the MCD method. The MCD estimation points identify a subset of observations with the smallest determinant of covariance and use this subset to calculate the result of a covariance matrix that is less affected by outliers. This method extremely improved the robustness of the analysis we did and reduces the risk of biased parameter estimates. Therefore, the robust covariance matrix serves as the foundation for the RCCA conducted in this study.

3.9. Canonical Correlation Analysis

RCCA was carried out to explore the multivariate relationship between digital distraction and deep learning approach by using the MCD approach. RCCA was selected because both constructs were represented by multiple variables which may interact at the same time. The analysis generates canonical variates, which are linear combinations of variables within each set that will maximize the correlation between the two constructs. This may result in canonical correlation coefficients which may indicate the strength of the relationship between the canonical variates, while canonical loadings provide information about the contribution of each original variable. These gauges were used to identify the dimensions that were the most strongly associated with the relationship between both digital distractions and deep learning approaches.

3.10. Interpretation of Results

Our results are interpreted based on:

- Canonical correlation values
- Canonical loadings
- Contribution of each variable

In addition, canonical loadings were examined to determine how strong each variable was associated with its own corresponding canonical function.Variables with larger absolute loading values were considered to contribute more substantially to the canonical relationship than the smaller absolute loading values variables.

4. RESULT AND DISCUSSION

4.1. Validity and Reliability Check

Before looking at the main relationships, the questionnaire itself needed to be tested for both Validity and Reliability. A validity check was used to make sure each of the 24 questions truly measured the right topic. Using Pearson's correlation, the results are:

TABLE I. Validity Check

Main Variable	Dimension	Item	R - Value	p-value	Status
Digital Distraction (X)	Social Media Urge (DD_D1)	x1	0.909	0.0000	Valid
		x2	0.871	0.0000	Valid
		x3	0.901	0.0000	Valid
		x4	0.903	0.0000	Valid
	Digital Notification Sensitivity (DD_D2)	x5	0.895	0.0000	Valid
		x6	0.934	0.0000	Valid
		x7	0.908	0.0000	Valid
		x8	0.926	0.0000	Valid
	Attention Fragmentation (DD_D3)	x9	0.887	0.0000	Valid
		x10	0.900	0.0000	Valid
		x11	0.900	0.0000	Valid
		x12	0.904	0.0000	Valid
Deep Learning Approach (Y)	Critical Thinking (DLA_D1)	x13	0.853	0.0000	Valid
		x14	0.915	0.0000	Valid
		x15	0.857	0.0000	Valid
		x16	0.862	0.0000	Valid
	Conceptual Understanding (DLA_D2)	x17	0.907	0.0000	Valid
		x18	0.868	0.0000	Valid
		x19	0.842	0.0000	Valid

		x20	0.841	0.0000	Valid
	Reflective Learning (DLA_D3)	x21	0.897	0.0000	Valid
		x22	0.860	0.0000	Valid
		x23	0.902	0.0000	Valid
		x24	0.920	0.0000	Valid

Based on the table above, each question (item) has a strong score (ranging from 0.841 to 0.934) and all of them are highly significant (p-value < 0.05). This is proof that the questions are relevant or valid towards their respective dimensions and main variables. Next, a reliability test using the Alpha Cronbach Formula was used to check if the questionnaire was consistent. The results are:

TABLE II. Reliability Check

Dimension	Result
Social Media Urge	0.917222739
Digital Notification Sensitivity	0.935775776
Attention Fragmentation	0.919721886
Critical Thinking	0.894132104
Conceptual Understanding	0.885945271
Reflective Learning	0.916865355

A score of 0.7 is commonly used as the benchmark for an acceptable and reliable questionnaire. The results showed that the measurement instrument exceeded this goal entirely. For the Digital Distraction categories, the reliability scores were 0.917 for DD_D1, 0.936 for DD_D2, and 0.920 for DD_D3 as well as 0.894 for DLA_D1, 0.885 for DLA_D2, and 0.916 for DLA_D3. Since all the dimensions yielded results above the benchmark value, the questionnaire is also proven to be highly reliable.

4.2. Multivariate Normality Evaluation

Many standard models in statistics require the data to follow a smooth, bell-shaped curve to fulfil the normally distributed assumption. To see if the respondents answers towards the questionnaire met said assumption, Mardia's test will be used since it's a multivariate research by evaluating the data's skewness and kurtosis.

TABLE III. Mardia's Test Results

Main Variable	Test	Statistic	p-value	Method	MVN
Digital Distraction	Mardia Skewness	27.315	0.002	asymptotic	Not Normal
	Mardia Kurtosis	2.534	0.011	asymptotic	Not Normal
Deep Learning Approach	Mardia Skewness	80.506	<0.01	asymptotic	Not Normal
	Mardia Kurtosis	6.746	<0.01	asymptotic	Not Normal

The Mardia test results above confirm that the dataset failed to meet the assumption of multivariate normality. Notably, prior testing with a smaller data sample size (n = 42) originally indicated a normal distribution. Having said that, as the data sample size went up to the full 107 respondents, significant skewness and kurtosis appeared. This change shows that the larger, more representative sample represented a wider fact of extreme behavioral variations, for example, a concentrated cluster of highly distracted university students, rather than clustering neatly around a standard average. Because standard statistical models mathematically assume a normal distribution, applying them to this full size dataset would achieve biased and inaccurate estimate results. Therefore, this shift in the data distribution strictly controls the need of pivoting to a robust statistical method that is free of distribution requirements.

4.3. Multicollinearity Check

It was also necessary to make sure the different parts of the questionnaire were not similar to each other, and in order to check this, a multicollinearity test was initiated. The overlap between the different dimensions was tested to ensure that the correlation scores stayed below the limit of 0.85 and 0.90 by using their average value as the base. The table below shows those scores:

TABLE IV. Correlation Matrix for Digital Distraction

	avg_dd_d1	avg_dd_d2	avg_dd_d3
avg_dd_d1	1	0.5974076	0.6332158
avg_dd_d2	0.5974076	1	0.6355840
avg_dd_d3	0.6332158	0.6355840	1

TABLE V. Correlation Matrix for Deep Learning Approach

	avg_dla_d1	avg_dla_d2	avg_dla_d3
avg_dla_d1	1	0.7781210	0.5279077
avg_dla_d2	0.7781210	1	0.5335725
avg_dla_d3	0.5279077	0.5335725	1

The correlation matrix above shows that multicollinearity is not present within the respective dimensions. In multivariate models, variables that have correlation scores above 0.85 (Kline, 2015) or 0.90 (Hair et al., 2018) shows redundancy, which can misrepresent the calculations of the model by double counting a single behaviour. But because all the observed correlations are below said thresholds, it is statistically established that each dimension measures a different concept. This makes sures that every dimension adds an independent variance to the final model without inflating the results.

4.4. Robust Canonical Correlation Analysis (MCD Approach)

Because the data violated the normality assumption, using a standard Canonical Correlation Analysis approach would yield estimates that are biased and heavily skewed by abnormal responses. To tackle this issue, a more robust method using Robust Canonical Correlation Analysis (RCCA) was used by utilizing the Minimum Covariance Determinant (MCD) approach via the robustbase library in R. Prior to the calculations, the dimensions were standardized to guarantee equal scaling. The fast MCD algorithm was used with an alpha parameter of 0.5, which successfully isolated a highly robust subset of 57 core samples. These samples are then utilized to calculate the robust location and covariance estimates, effectively nullifying the presence and influence of extreme outliers.

TABLE VI. Robust Estimate of Covariance

	avg_dd_d1	avg_dd_d2	avg_dd_d3	avg_dla_d1	avg_dla_d2	avg_dla_d3
avg_dd_d1	0.7198	0.4798	0.4831	0.2088	0.2514	0.3430
avg_dd_d2	0.4798	0.7344	0.5870	0.2498	0.2180	0.4726
avg_dd_d3	0.4831	0.5870	0.6133	0.2455	0.2099	0.4667
avg_dla_d1	0.2088	0.2498	0.2455	0.5669	0.5704	0.4237
avg_dla_d2	0.2514	0.2180	0.2099	0.5704	0.7815	0.4550
avg_dla_d3	0.3430	0.4726	0.4667	0.4237	0.4550	0.9379

Standard mathematical methods compute relationships using all data points equally, making them highly sensitive towards extreme outliers. The MCD approach corrects this thus providing an unbiased estimation of the core behavioral pattern within the sample.

4.5. Canonical Functions

The robust formula searched for mathematical connections that link digital distractions to deep learning approaches. These are called canonical functions in multivariate statistics. The goal is to find the strongest possible link between these two main variables. The model calculates three possible connections with correlation scores and shared variance percentages for each connection were calculated directly from the eigenvalues of the robust matrix.

TABLE VII. Canonical Correlation Coefficients and Shared Variance

Function	**Canonical Correlation**	**Shared Variance (Eigenvalues)**
1	0.647646030	41.94%
2	0.266804001	7.12%
3	0.002012803	<0.01%

The robust analysis above shows proof that there is a strong relationship involving multiple dimensions between digital distraction and deep learning approach. This relationship is mainly captured by the first canonical function, which represents the strongest and most successful blend of the data. The shared variance score of 41.94% means that a substantial portion (~42%) of the changes seen in a university student deep learning approach can be statistically explained by looking at the digital distraction portrayal. On the other hand, the other two canonical functions are extremely low, which proves that the structural relationship works almost entirely through only one main connection, rather than being scattered across many different factors.

4.6. Variable Loadings

Because the first canonical function resulting from before holds the core behavioral pattern, it becomes necessary to examine the canonical loadings for this specific function. The purpose of canonical loadings is to measure the direct linear correlation between the original questionnaire dimensions and the newly created variables. In practical words, the loadings show how much weight each dimension carries inside that function. This effectively creates a ranking system to show the impact each dimension provides.

TABLE VIII. Canonical Variable Loadings for Function 1 (X)

Social Media	Digital Notification Sensitivity	Attention Fragmentation
0.5075	0.6946	0.9781

TABLE IX. Canonical Variable Loadings for Function 1 (Y)

Critical Thinking	Conceptual Understanding	Reflective Learning
0.6883	0.4148	0.9039

Each dimension's specific contribution to the overall structural relationship is indicated by the canonical loadings. According to Table VIII, attention fragmentation is the primary indicator in the digital distraction group, as shown by its strong loading score of 0.9781. This suggests that the most significant component of the overall characteristic of digital distraction is attention fragmentation. However, Table IX indicates that, with a loading value of 0.9039, Reflective Learning is the deep learning approach's most impacted outcome characteristic. These findings demonstrate that certain factors influence the relationship between technology and academic focus. The statistical model demonstrates that a significant level of attention fragmentation is directly related to the reflective learning dimension, which produced the greatest canonical loading outcome in the model, contradicting the original negative hypothesis.

5. CONCLUSION

This research successfully examined the multivariate relationship between digital distraction and deep learning approach using a robust statistical method. By using the Minimum Covariance Determinant approach, the analysis managed to nullify extreme outliers and provided a highly reliable estimation for the core samples. The analysis confirms that there is a relationship between the two constructs / main variables, directly answering the problem foundations. Specifically, the first canonical function demonstrates that the overall digital distraction profile can statistically explain 41.94% of the variance in the deep learning approach.

Additionally, the results of the variable loadings revealed the most important variables in this connection. The main source of digital distraction is attention fragmentation, yet reflective learning yielded the greatest loading value, indicating that students believe they can think deeply even in the face of digital disruptions. These findings demonstrate that modern digital disruptions not only lead to general academic tiredness but can also skew students' perceptions of their own ability to think deeply and thoughtfully. In practical terms, these results highlight the necessity for teachers to create learning environments that reduce digital multitasking while also motivating students to strengthen their self-regulated learning habits and metacognitive awareness in order to accurately assess and maintain their cognitive focus.

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